Optiver

Optibook Guide

FutureFocus 2026

1 Introduction

Optibook is a fully functional virtual exchange, built in-house at Optiver for educational purposes. Through Optibook, we teach about exchanges, financial instruments and market making, and aim to show why market makers are essential for the functioning of financial markets.

As part of FutureFocus 2026, you will develop an automated trading algorithm (or “autotrader”) to trade on the Optibook exchange and generate as much profit as possible. The Instrument reference section of this document contains an overview of all the tradable products listed on the exchange; your algorithm may trade as many of these products as you wish.

This exercise does not have a single correct answer; instead, it is open-ended, much like the challenges we often face on the trading floor. As in real financial markets, your algorithm will have to deal with the uncertain and unpredictable behaviour of other market participants, ideally adjusting to the situation at hand. Rather than trying to derive the optimal algorithm once-and-for-all, the algorithms should be built up by a combination of first principles and what-if style thinking. The algorithm you develop will be an engineering project, with many moving parts, all working in tandem. Choices you make in one part will relate to what the optimal choices in another are.

You are expected to take full ownership of the design and implementation of your algorithm. You may use AI tools (e.g. code assistants, large language models) as part of your workflow; however, you remain fully responsible for the strategy you submit. You must understand every component of your algorithm and be able to clearly explain your approach, assumptions, strategic edge, risk management, and implementation decisions to a third party. Reliance on generated code without a deep understanding of its logic or trade-offs will not be sufficient.

While light collaboration with other teams on minor issues (e.g., environment setup or syntax questions) is acceptable, the core strategy, modelling choices, and overall system design must be your own work. Additionally, do not share your output publicly, and do not download data from the Optibook server to your personal devices.

On your first day, there will be a lecture to introduce you to Optibook and its features. Additional reference materials, (such as recorded lectures of financial concepts, API documentation, etc) are available on the Optibook webpage hosted at https://2026-syd-ff2.optibook.net/ under docs and resources.

2 Programming environment

Algorithms that trade on Optibook are written in Python. Python is an interpreted programming language. This means that a Python program is defined in a text file, which is read and executed by a Python interpreter.

To avoid any differences between setups, we have opted for you to write and run your code through a cloud-based setup hosted on Amazon Web Services (AWS). What this provides is an environment that is correctly set up from the first time you launch it, offering a pre-installed version of Python, a browser-based IDE (Visual Studio) supporting .py files and Jupyter notebooks, and the ability for us to preload the Optibook libraries and any example code files. For simplicity, we ask that you use this web-hosted service from a personal laptop. It is highly recommended to use a Chrome browser for access.

To sign up to the platform:

1. Browse to https://2026-syd-ff2.optibook.net/
2. Click on the **Sign Up** button on the top-right of the page
3. Enter your email address (the one you used for Optiver's online assessment portal) as well as your full name
4. Choose and confirm a password
5. Click **SIGN UP.** If your email is correct, you should see a green box: "Account created successfully"
    - If you get an error at this step, it's likely you used a different email on our online assessment portal. Reach out to Marcus if you need help identifying which email you should be using.

To access your development environment (devenv):

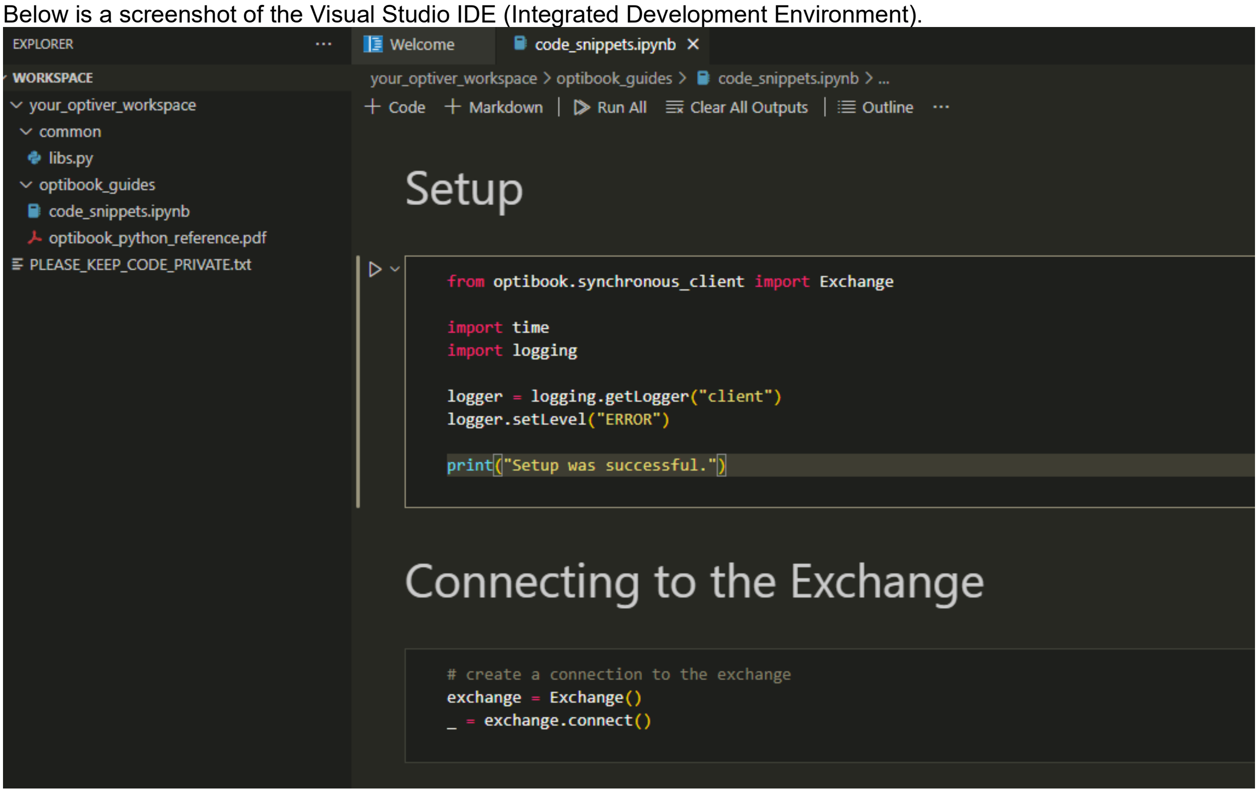
1. Browse to https://2026-syd-ff2.optibook.net/
2. Click on the **Login** button on the top-right of the page
3. Enter your email address and password from the previous steps
4. Click on the green Dev Environment button
5. Check the box indicating that you trust the authors of all files in the parent folder (one time only)

Troubleshooting common errors if you are unable to login:

- You browsed to the wrong server URL (e.g. 2026-syd-ff2**-team-XXX**.optibook.net instead of 2026-syd-ff2.optibook.net)
- You typed your password incorrectly, for example by including a trailing space

2.1 Visual Studio Usage

Below is a screenshot of the Visual Studio IDE (Integrated Development Environment).



The large area in the top-right is where text and code are edited. If the tab that is currently selected in the main area is a .py file, it can be run directly by hitting the "play" button located at the top of the screen. The output of the file is then displayed at the bottom of the screen.

The column on the left is a file browser, which should contain some introductory code snippets and guides. Double clicking a file opens it as a tab in the code editor. Feel free to make use of any other functionality supported in the IDE.

2.2 Jupyter notebooks

Jupyter notebooks are supported directly from the IDE. The '*code_snippets.ipynb'* notebook is a good starting point; it contains examples of many interactions that are possible with the Optibook exchange through the Exchange object. When first running a notebook, you must configure which version of Python (kernel) is used. To do so:

- Open a Jupyter notebook
- Wait until the loading icon on the top right of the notebook, *'Loading kernels...'*, is replaced by *'Select Kernel'*
- Click *'Select Kernel'*
- Above the notebook, in the center, there is now a list of options to select. Select *'Python environments...'* followed by the recommended option

Your notebook is now configured correctly and you're able to run the cells as normal. Visual Studio should remember which Kernel/Python version you selected for any future notebooks you open - but if not, simply repeat the above.

3 Instrument reference

This section contains a full reference of the products available for trading. On the Optibook Visualizer, the Instrument panel shows all listed instruments, i.e. all tradeable products. We refer to these instruments by their unique **instrument_id**, shown in the first column of the Instrument Overview. The **instrument_id**'s can be interpreted to understand what specific instrument or derivative is being referred to (for example, the expiry date of a future contract). But this information can also be obtained directly through code. An example of how to do so is provided in the *'code_snippets.ipynb'* notebook in your workspace. Note that some code in *'code_snippets.ipynb'* might reference instrument_id's that aren't applicable for FutureFocus 2026 (e.g. 'AAPL').

The tick size for all products is 0.10 and there are no exchange fees.

3.1 Stocks

The **instrument_id** of a stock is simply its ticker, e.g. 'NVDA', 'JPM', 'XOM', etc. Stocks are cash instruments (i.e. not derivatives) representing a unit of ownership in a company. As is the case for all instruments on Optibook, the stocks can be **sold short** i.e. one is allowed to have a **negative** position in them. The real-world process of selling a stock short is an involved process whereby one borrows the stock from someone else first to return later. On Optibook, this is all fortunately abstracted away; you can simply enter into a negative position, and this will generate the opposite profit-and-loss (PnL) as a positive (i.e. long) position would. The names and prices of these fictitious stocks have no relation to any real-world companies, and none of the stocks pay any dividends.

3.2 Stock dual listings

A *dual listing* is the listing of any security on two different exchanges. As they represent the exact same rights of ownership, it follows that they should have the same value, but the available prices can deviate at times due to the disjoint nature of the different exchanges. Our simulated exchange is a single exchange in a single location, listing different instruments – however, to simulate dual listings, we introduce a second instrument on some of the stocks, with the instrument_id **<stock_ticker>_DUAL**. These dual listings are very similar to the dual listings on real exchanges, in that they represent the same stock but have disjoint order books, leading to interesting dynamics between them.

For dual listings on real exchanges, it is usually possible to transfer a stock holding from one exchange to the other. This ability completes the arbitrage, so to speak, as one might buy a stock on one exchange, transfer it, and sell it at the other. Such a transfer is not possible for you on Optibook, but you may assume that it is possible for the broader market. For the purposes of exchange-enforced limits, Optibook will consider a dual listing as two separate stocks.

3.3 Index

Individual stocks are often combined into a basket to create a *stock index*. An *index value* is calculated as the weighted average of the constituent stock values, for some pre-decided weighting scheme which can vary per index. Tracking the value of these indices allows investors to focus on broad market developments, e.g. in specific sectors or countries, rather than the idiosyncrasies of each individual company's performance. Some examples of well-known financial indices are the S&P 500, NASDAQ, and the ASX 200.

<u>Indices themselves are not tradeable directly</u>. The index value is simply a number that is defined, calculated and officially published by some financial institution, often an exchange. For investors, it is possible to get a direct monetary exposure to this number either by buying the constituent stocks or by trading an index-derived product.

Optibook defines one stock index, the OB5X. It is a value-weighted index with the following weighting scheme:

$$X_t = \frac{\sum_{i=1}^{5} w_i \cdot S_{i,t}}{1000}$$

Here X_t represents the index value at time t, w_i represents the index weight of stock price i based on the fictional weighting scheme, $S_{i,t}$ represents stock i's value at time t, and 1000 is a normalising constant. For simplicity, the weighting factors w_i are kept constant throughout the simulation at the following values:

Company	AMZN	JPM	NVDA	XOM	NVO
w_i	953.21	129.25	908.06	2245.39	124.78

For dual-listed stocks, only the non-DUAL stock price is considered in calculating the level of the index.

To reiterate, OB5X itself is not a tradeable instrument, and Optibook does not explicitly publish its value.

3.4 Index ETF

An ETF is an exchange-traded fund. It can be thought of as the stock of a publicly listed company which is incorporated for a single purpose: to hold balance sheet assets (here, stocks) in the appropriate ratios to track the performance of a pre-defined financial index.

The issue with manually tracking an index like the S&P 500 is that one needs to buy and hold all the 500 different stocks in the appropriate ratios. This is firstly quite cumbersome and inefficient; one incurs trading costs for each individual transaction in the stocks and would need to rebalance continuously as the ratios change. Secondly, it is not possible to hold fractional stocks. If one would need to hold two stocks in a ratio of 30:1, there is no other option than to buy 30 of the second stock for every 1 of the first, thereby requiring a large total exposure. These issues are resolved through the concept of an ETF. The ETF issuer manages the trading and rebalancing of the stocks in the correct ratios, and the interested market participant can get exposure to the index through a single trade in the ETF itself.

Since the ETF needs to rebalance periodically, it can sometimes be left with a small cash position in addition to the stocks it holds, representing the money it has left over from its transactions. Therefore, we can describe the theoretical value of the ETF with the following relationship:

$$V_t = NAV_t = C_t + M \cdot X_t$$

Here NAV_t stands for the net-asset value at time t, C_t represents the (potentially time-varying) cash component of the fund, M represents a fixed multiplier consistent with the number of outstanding shares relative to the ETF holdings, and X_t represents the time-t index value. In other words, it's simply worth the sum of all of its assets, including the cash holding and replicated index exposure. Optibook lists an ETF on OB5X with instrument_id **OB5X_ETF**; it has multiplier M = 0.25 and a time-constant cash component $C_t = C$ = 2.50.

Optiver

3.5 Index futures

Index futures are financial derivatives that allow traders to speculate on or hedge against the future value of an index. Unlike some other types of futures, index futures do not involve the physical delivery of assets. Instead, they are cash-settled, meaning that at expiration, the difference between the traded price and the actual index value is paid or received in cash. Procedures like “margining” and “netting”, which relate to the exchange’s management of the available funds of derivatives traders, are fortunately abstracted away on Optibook.

Optibook lists OB5X futures on the nearest quarterly expiries (by convention, the expiry always takes place on the third Friday of the month at 12:00 UTC). Their instrument_ids have the form **OB5X_<yyyy><mm>_F.** The theoretical value of these futures contracts can be derived directly from the following formula: $F_t = X_t \cdot \exp(r\tau)$ where X_t is the index value at time t, annualized interest rate $r = 0.03$, and τ is time until expiry in years. The contract size is 1x the OB5X index.

4 Exchange-enforced limits

The Optibook exchange imposes several exchange-side limits that your algorithm must adhere to:

- Total position (long or short) per instrument cannot exceed 100.
- Total orders outstanding per instrument cannot exceed 200.
- You can send no more than 25 updates (inserts/deletes/amends) per second.

These limits are applied pre-emptively. For an inserted order, if there is any worst-case scenario of order execution conceivable, by matching with any current and/or future opposing orders, that may lead to breaching a limit, the order is rejected by the exchange, and the participant/algorithm directly disconnected. On disconnect, all outstanding orders are deleted. While you may simply reconnect, it is best to stay in full control of your algorithm to avoid breaching these limits in the first place. Insert timers to avoid breaching the frequency limit, and ensure your outstanding order volume is capped to a value that does not cause breaches of the position and outstanding order volume limits.

5 Trading hours and evaluation

The Optibook exchange will be open during the designated Optibook blocks in your FutureFocus 2026 schedule. Outside of these hours, trading will be halted and your positions and PnLs reset. We ask that you do not perform any work on this exercise after hours.

Evaluation of your trading algorithm will take place on Friday during a 30 minute trading window. At the end of this window, expiration procedures will be run to determine your final PnL: the 5 stocks will be expired at their last traded prices, the DUAL listings will be expired at the last traded price of their non-DUAL counterparts, and the ETF and index futures will be expired at their theoretical values.

We can’t wait to see what you build. Happy trading!